

Course Code	Problem Solving with C	L	T	P	C

COURSE DESCRIPTION

This course introduces the fundamentals of programming using the C language. Students learn structured problem solving, algorithmic thinking, and core programming constructs including control structures, arrays, strings, pointers, functions, structures, and file handling. The course emphasizes hands-on implementation and builds a strong foundation for advanced programming, data structures, and systems courses.

COURSE OBJECTIVES

The course should enable the student to:

1. Explain the fundamentals of the C language, program structure, and execution model.
2. Apply operators, decision-making statements, and looping constructs to solve computational problems.
3. Develop programs using arrays, strings, and pointers for structured data handling.
4. Design modular programs using functions, recursion, and dynamic memory allocation techniques.
5. Implement structures and file handling concepts in practical programming applications.

SYLLABUS

Unit 1: Fundamentals of C Programming

What is a Computer Program, Why C Language, Structure of a C Program, Compilation and Execution Process, GCC Compilation, Variables, Data Types, Constants, Arithmetic Operators, Assignment Operators, Relational Operators, Logical Operators, sizeof Operator, Type Casting, printf, scanf

Unit 2: Decision Making and Looping Constructs

if Statement, if-else Statement, if-else-if Ladder, Nested if, switch-case Statement, Ternary Operator, while Loop, for Loop, do-while Loop, Nested Loops, break, continue

Unit 3: Functions, Arrays and Strings

Function Declaration and Definition, Function Prototypes, Parameter Passing, Return Values, Scope and Storage Classes, One-Dimensional Arrays, Two-Dimensional Arrays, Passing Arrays to Functions, Character Arrays, Strings, String Library Functions

Unit 4: Pointers and Dynamic Memory Management

Pointers, Address and Dereferencing Operators, Pointer Arithmetic, Arrays and Pointers, Pointers and Functions, Recursion, Dynamic Memory Allocation

Unit 5: Structures and File Handling

Structures, Nested Structures, Arrays of Structures, Union, enum, typedef, File Handling, Text Files, Binary Files, File Positioning, Command Line Arguments

TEXTBOOKS/REFERENCE BOOKS

TEXTBOOKS

- **T1:** The C Programming Language – Brian W. Kernighan & Dennis Ritchie
- **T2:** Programming in ANSI C – E. Balagurusamy

REFERENCE BOOKS

- **R1:** Let Us C – Yashavant Kanetkar
- **R2:** C Programming: A Modern Approach – K.N. King

PRACTICALS

Experiment 1: Develop a C program for a simple Electricity Bill Calculator that computes total bill amount based on slab-wise tariff rates using formatted input and output.

Experiment 2: Develop a C program to generate a number pyramid and Floyd's triangle pattern using nested loops.

Experiment 3: Write a C program to analyze student marks stored in an array and compute highest, lowest, average and grade distribution.

Experiment 4: Develop a C program to perform matrix multiplication and transpose for two matrices entered by the user.

Experiment 5: Write a C program to check whether a given string is a palindrome and count vowels, consonants, digits and spaces without using built-in string functions.

Experiment 6: Write a C program to swap two numbers using pointers, and demonstrate pointer arithmetic on an integer array.

Experiment 7: Develop a C program to dynamically allocate memory for n numbers using malloc, then find the second largest element in the list.

Experiment 8: Write a modular C program using functions to implement a Bank Account System supporting deposit, withdrawal and balance inquiry.

Experiment 9: Develop a recursive C program to solve the Tower of Hanoi problem and display the sequence of moves.

Experiment 10: Create a Student Record Management System using structures and file handling to add, display and search student records stored in a file.